

CS3151 SOFTWARE ENGINEERING

LAB 4

From Personas & Requirements to Use Cases & UML

BodyBalance AI

Item	Information
Course	CS3151 Software Engineering
Project	BodyBalance AI
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Instructor	Dr. Passent Elkafrawy
Semester	Fall 2026
Deliverable	UML Lab - D1 to D7

1. Introduction

BodyBalance AI is a health, fitness, and recovery decision-support platform. It combines the user's physical and mental condition, symptoms, activity, recovery, and available BodyScope AI observations to provide personalized next-step guidance. Medical diagnosis and treatment remain the responsibility of qualified healthcare professionals.

To keep the UML readable, closely related lower-level functions are consolidated into five goal-level use cases. The course system-modeling material allows a complex use case to be expanded into its own detailed use case diagram; this report uses that approach instead of one crowded diagram.

System Context Overview (Supporting Model)

The context model shows the BodyBalance AI system boundary and only the external people that exchange information with the system. It is included as a quick orientation before the detailed use cases.

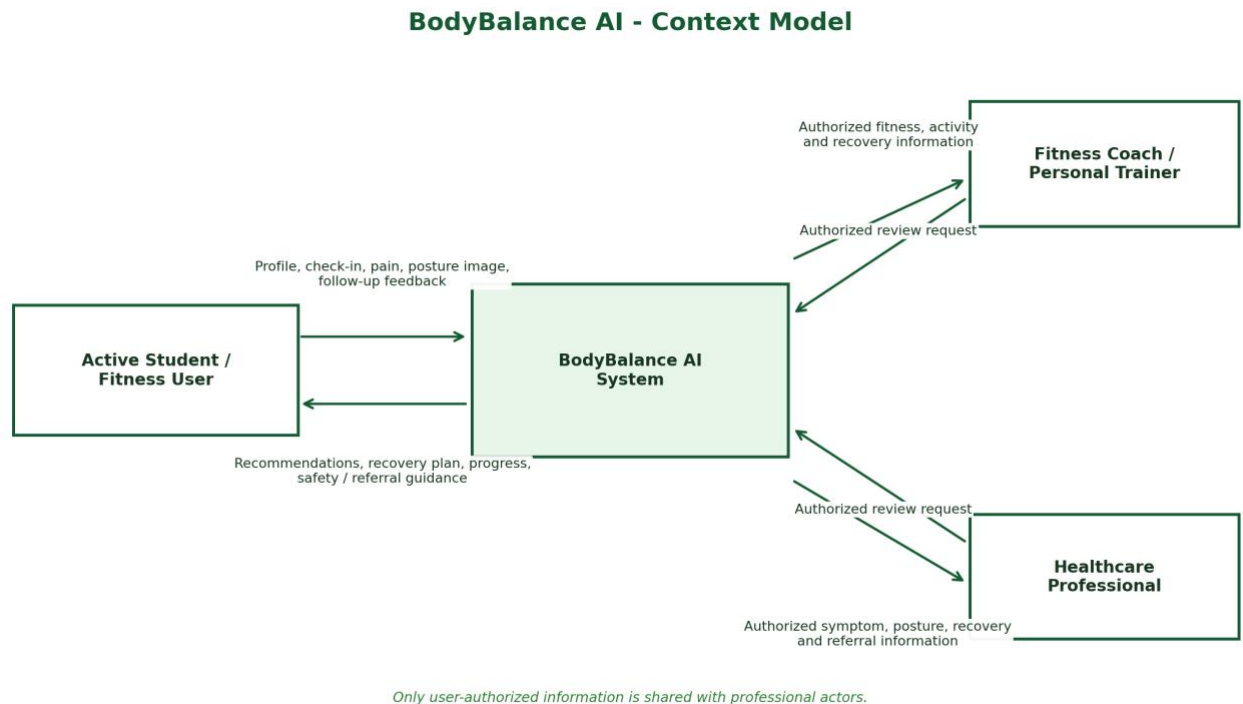


Figure 1. BodyBalance AI Context Model

D1. Actor Definitions and Prioritized Use Case List

D1.1 Actor Definitions

Actor	Description	Triggers / Interests	Challenges	Notes
Active Student / Fitness User	Main BodyBalance AI user who records physical, mental, pain, posture, activity, and recovery information.	Understand current condition, receive personalized guidance, monitor progress, and know when professional evaluation may be appropriate.	Pain, fatigue, stress, poor recovery, limitations, and uncertainty about exercise versus recovery.	Persona: Sara Ahmed.
Fitness Coach / Personal Trainer	Professional who reviews client-authorized fitness, activity, and recovery information.	Use relevant client information to adjust training recommendations appropriately.	Incomplete information and user-controlled access.	Persona: Omar Khalid.
Healthcare Professional	Authorized physiotherapist or clinician who reviews user-authorized symptom, activity, posture, and recovery information.	Use organized information to support professional assessment without replacing professional judgment.	Incomplete histories and the need for authorized, relevant information.	Persona: Ahmed Hassan.

D1.2 Consolidated Use Case List

ID	Use Case Name	Primary Actor	Goal / Outcome	Priority	Merged Functions
UC-01	Manage User Account	Active Student / Fitness User	Create, access, and maintain the user account and profile.	M	Register Account; Authenticate User; Manage Profile
UC-02	Perform Body Assessment	Active Student / Fitness User	Capture current physical/mental condition, pain, and supported posture observations.	M	Daily Check-In; Pain Assessment; Analyze Posture
UC-03	Generate Safe Activity Recommendation	Active Student / Fitness User	Perform safety screening and provide condition-aware activity guidance or referral when needed.	M	Safety Check; Activity Recommendation; conditional Referral Guidance
UC-04	Manage Recovery & Progress	Active Student / Fitness User	Generate recovery guidance, collect follow-up feedback, and review progress over time.	M	Recovery Plan; Follow-Up Feedback; Progress History
UC-05	Manage Professional Access	Active Student / Fitness User	Control authorized sharing and support professional review of permitted information.	S	Data Permissions; Professional Summary; Coach Review; Healthcare Review

Priority key: M = Must have; S = Should have; C = Could have; W = Won't have for the current scope.

D2-D5. Use Case Specifications and UML Views

Each consolidated use case is presented as one grouped UML section containing its detailed Use Case Diagram, Activity Diagram, and Sequence Diagram. UC-03 remains the most complex case required by the lab, while the additional activity and sequence diagrams are included for completeness and consistency across the five use cases.

UC-01 Manage User Account

Brief Specification (D2)

Field	Content
Primary Actor	Active Student / Fitness User
Goal	Create, access, and maintain an account.
Preconditions	BodyBalance AI is available.
Postconditions	Account/profile changes are securely stored.
Main Success Scenario	User registers an account or manages profile information; authentication is required before protected profile management.

Detailed Use Case Diagram (D3)

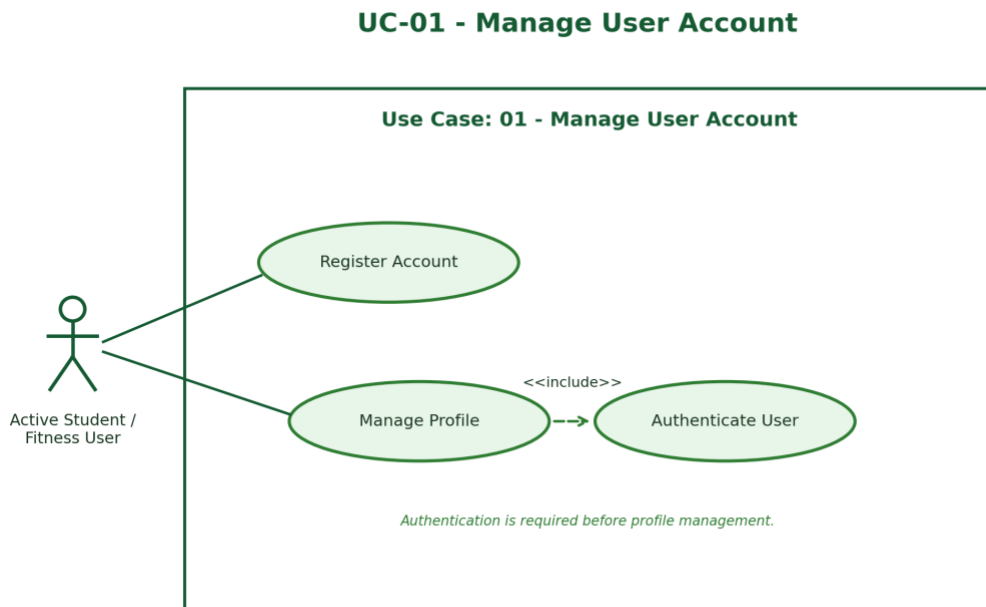
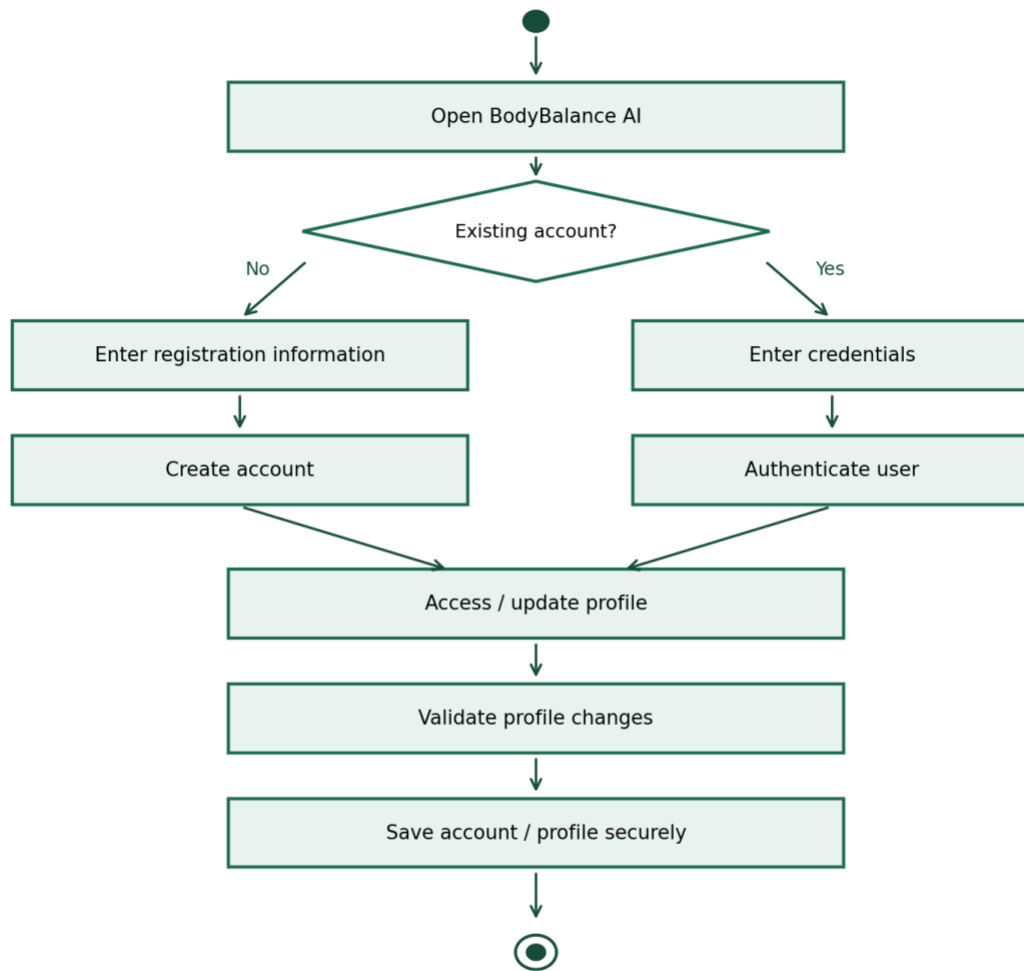


Figure 2. UC-01 Manage User Account - detailed use case diagram

Activity Diagram

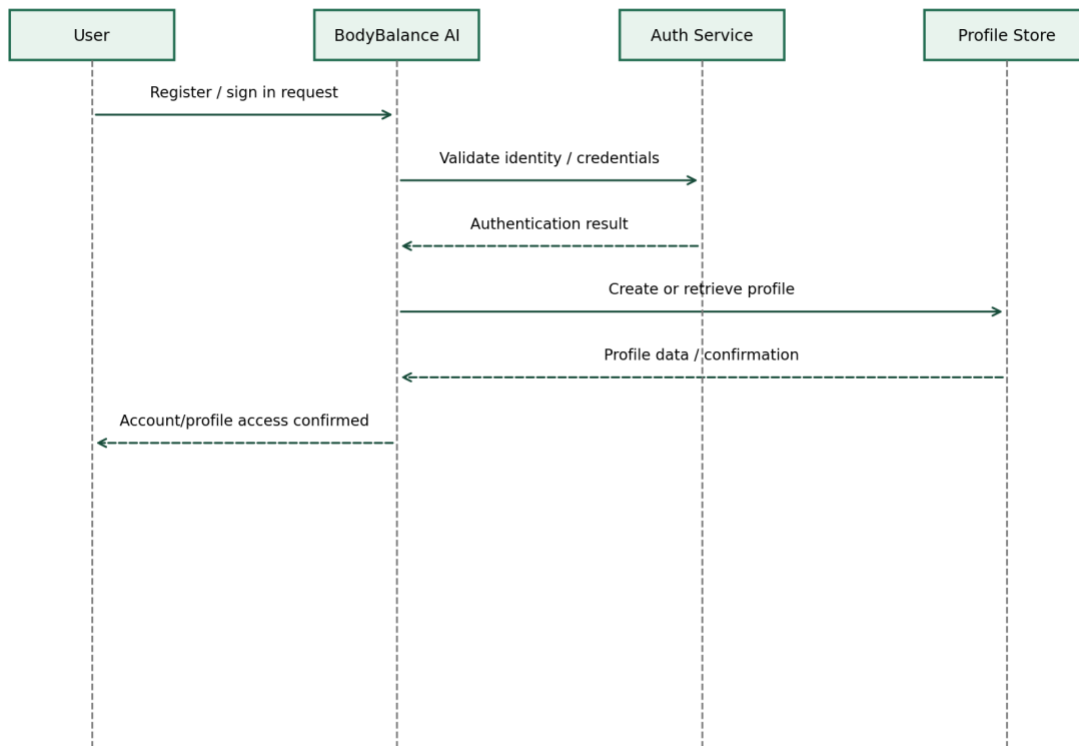
UC-01 Activity Diagram



UC-01 Activity Diagram

Sequence Diagram

UC-01 Sequence Diagram



UC-01 Sequence Diagram

UC-02 Perform Body Assessment

Brief Specification (D2)

Field	Content
Primary Actor	Active Student / Fitness User
Goal	Capture the user's current condition.
Preconditions	User is authenticated.
Postconditions	Current assessment information is validated and stored.
Main Success Scenario	User completes the daily check-in. Pain assessment is added when pain is reported, and posture analysis is added when a supported image is submitted.

Detailed Use Case Diagram (D3)

UC-02 - Perform Body Assessment

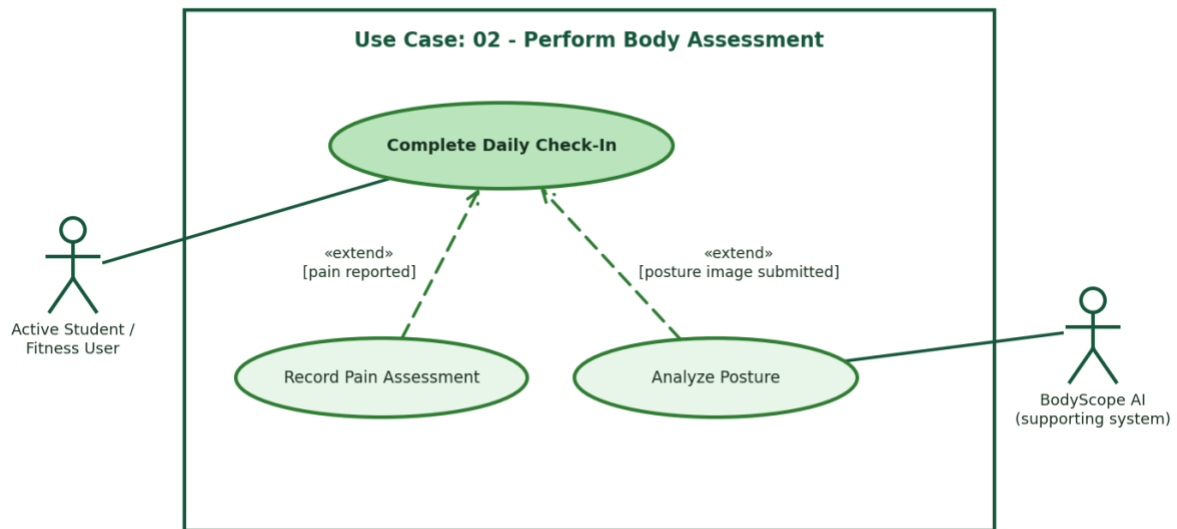
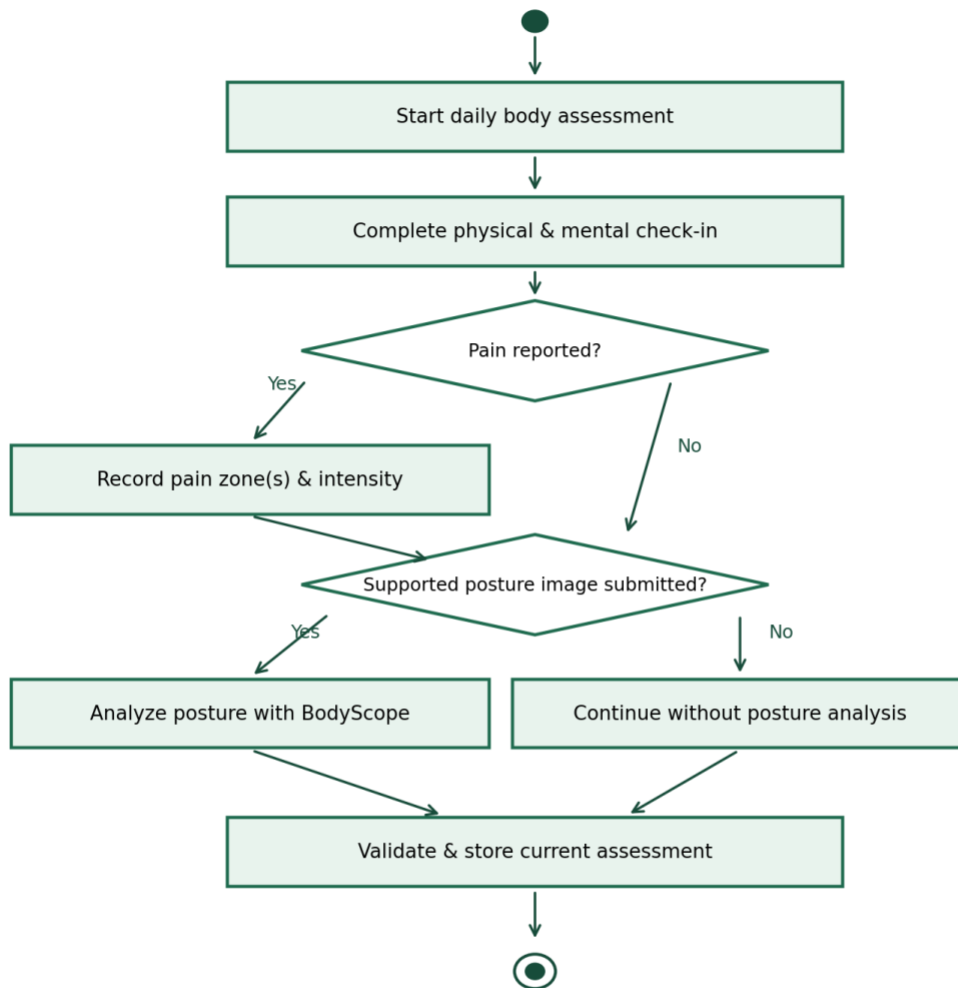


Figure 3. UC-02 Perform Body Assessment - detailed use case diagram

Activity Diagram

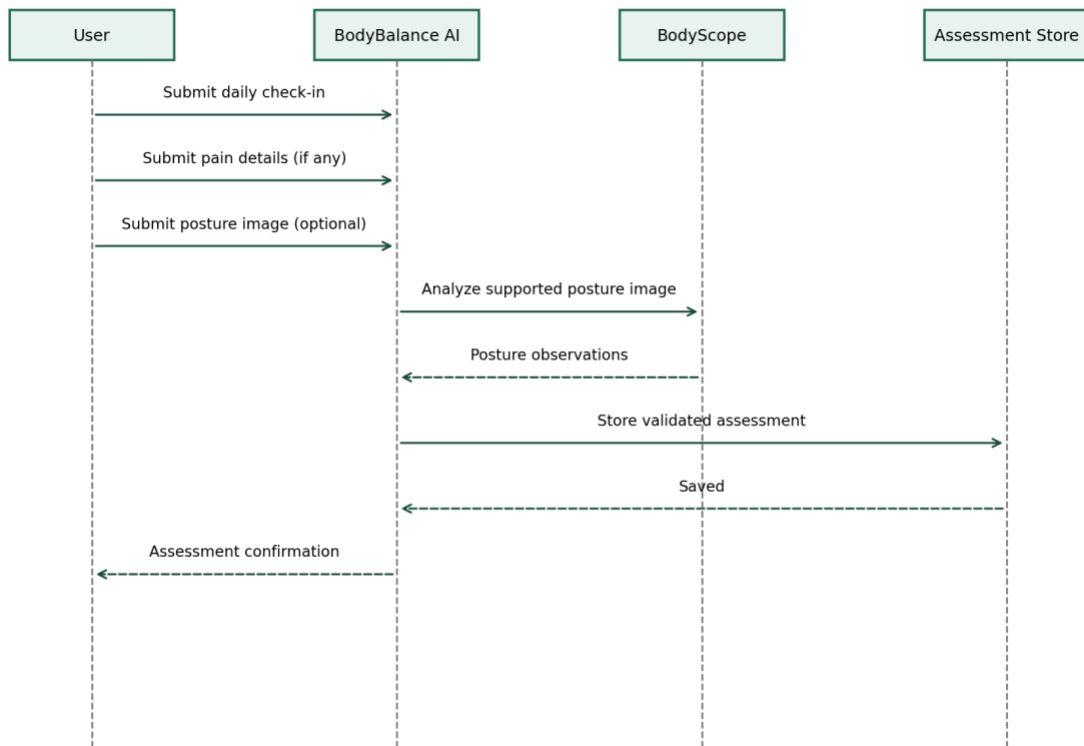
UC-02 Activity Diagram



UC-02 Activity Diagram

Sequence Diagram

UC-02 Sequence Diagram



UC-02 Sequence Diagram

UC-03 Generate Safe Activity Recommendation

Fully Dressed Specification (D2 - Core / Most Complex Use Case)

Field	Content
Use Case	UC-03 Generate Safe Activity Recommendation
Activity Name	Generate Safe Activity Recommendation
Primary Actor	Active Student / Fitness User
Stakeholders & Interests	User: wants safe, relevant guidance. Fitness Coach: wants recommendations to reflect relevant client information. Healthcare Professional: wants AI output to remain decision-support information.
Preconditions	User is authenticated; current assessment information is complete enough; applicable profile and assessment information can be retrieved.
Postconditions	A personalized recommendation is displayed with explanation. If a concerning indicator is detected, routine exercise guidance is blocked and referral guidance is provided.
Main Success Scenario	1. User submits current assessment. 2. System validates required information. 3. System performs safety check. 4. System retrieves applicable profile, check-in, pain, activity, and available BodyScope information. 5. Recommendation engine generates activity guidance. 6. System generates and displays the recommendation explanation. 7. User reviews the recommendation.

Extensions / Alternatives	2a. Missing information -> request completion before continuing. 3a. Concerning indicator -> block routine exercise guidance, display warning, and provide referral guidance. 4a. BodyScope unavailable -> use other applicable available information without inventing posture results.
Special Requirements	Main user-reported factors contributing to the recommendation shall be identified. AI output shall be presented as decision-support information rather than a confirmed medical diagnosis.
Constraints	Diagnosis and treatment remain outside the intended role of the system.
Open Issues	Exact recommendation logic and supported activity categories require later validation and testing.

Detailed Use Case Diagram (D3)

UC-03 - Generate Safe Activity Recommendation

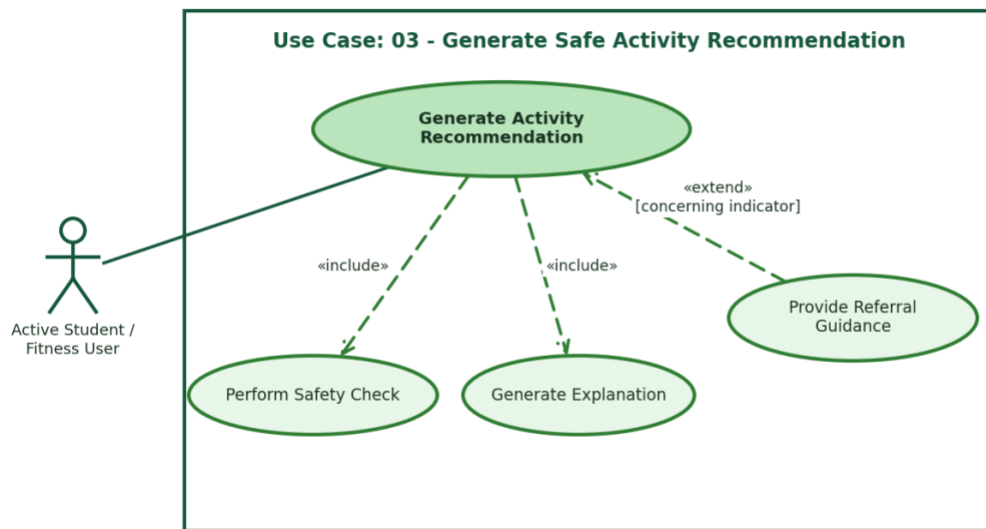


Figure 4. UC-03 Generate Safe Activity Recommendation - detailed use case diagram

Activity Diagram

UC-03 Activity Diagram - Safe Recommendation Workflow

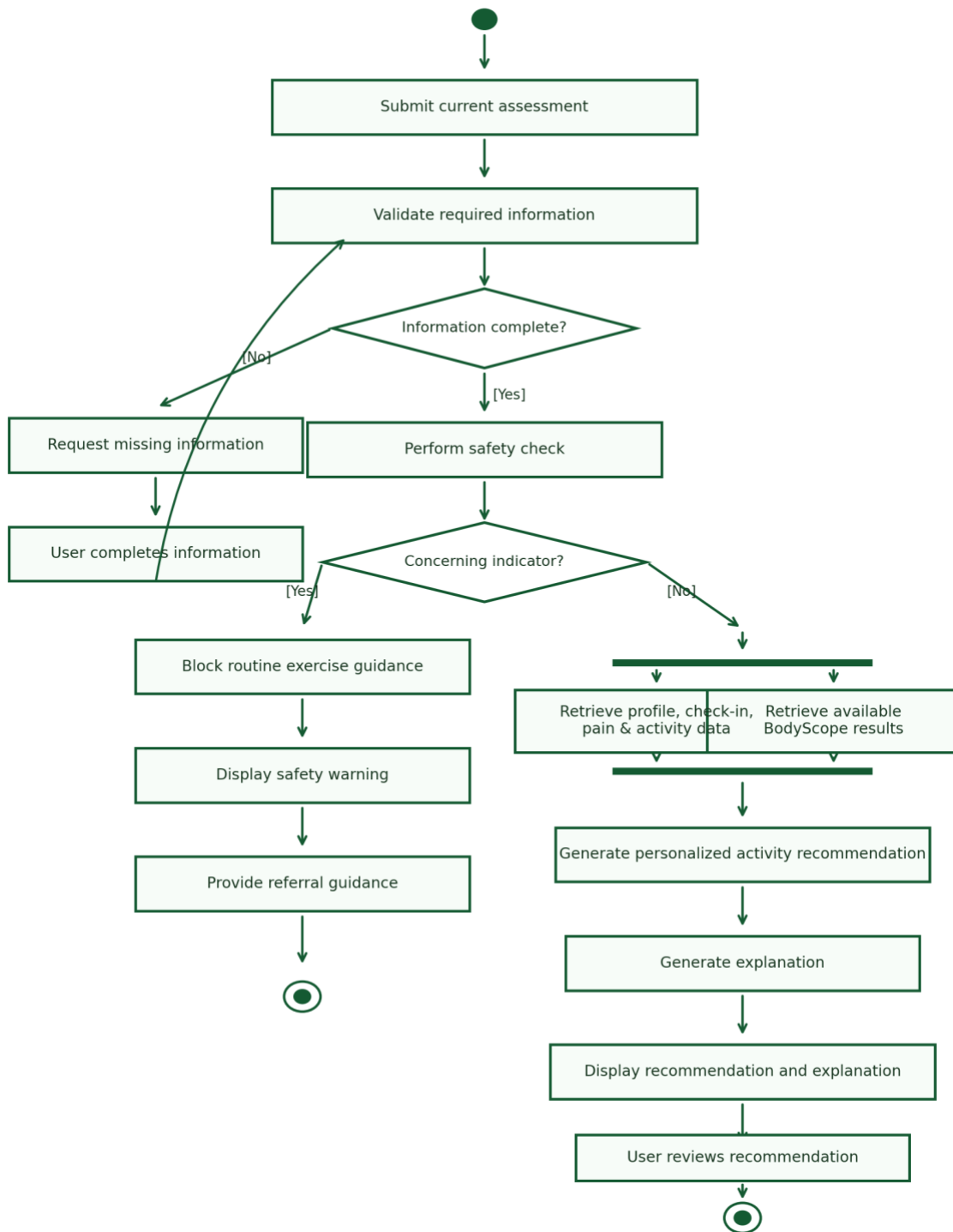


Figure 5. UC-03 Activity Diagram - safety check and recommendation workflow

Sequence Diagram

UC-03 Sequence Diagram - Main Success Scenario

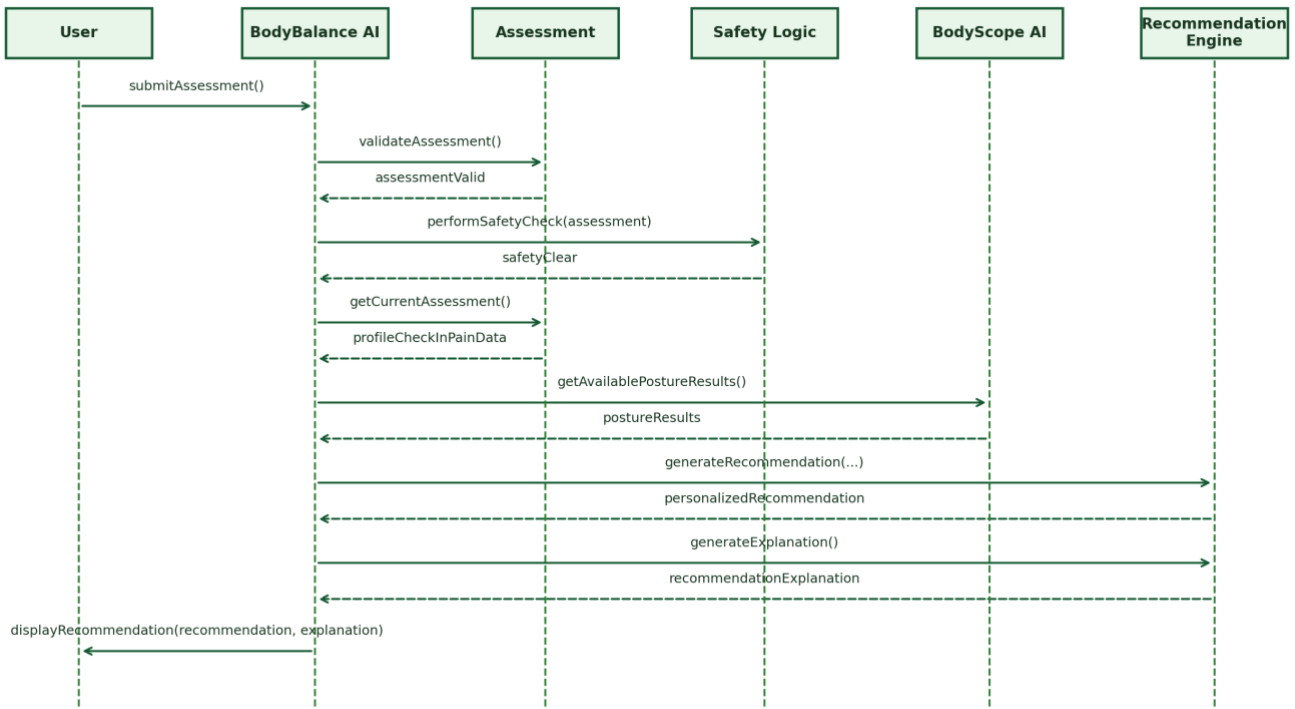


Figure 6. UC-03 Sequence Diagram - main success scenario

UC-04 Manage Recovery & Progress

Brief Specification (D2)

Field	Content
Primary Actor	Active Student / Fitness User
Goal	Support recovery and longitudinal progress.
Preconditions	Relevant assessment or recovery information exists.
Postconditions	Recovery guidance, follow-up feedback, and progress history are available.
Main Success Scenario	System generates recovery guidance; the user later submits follow-up feedback and reviews progress over time.

Detailed Use Case Diagram (D3)

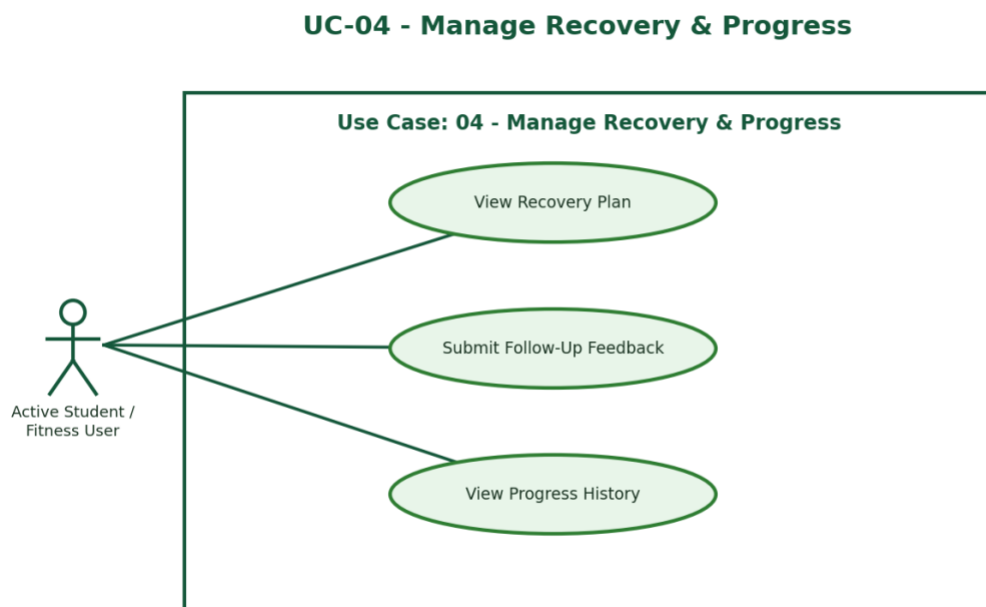
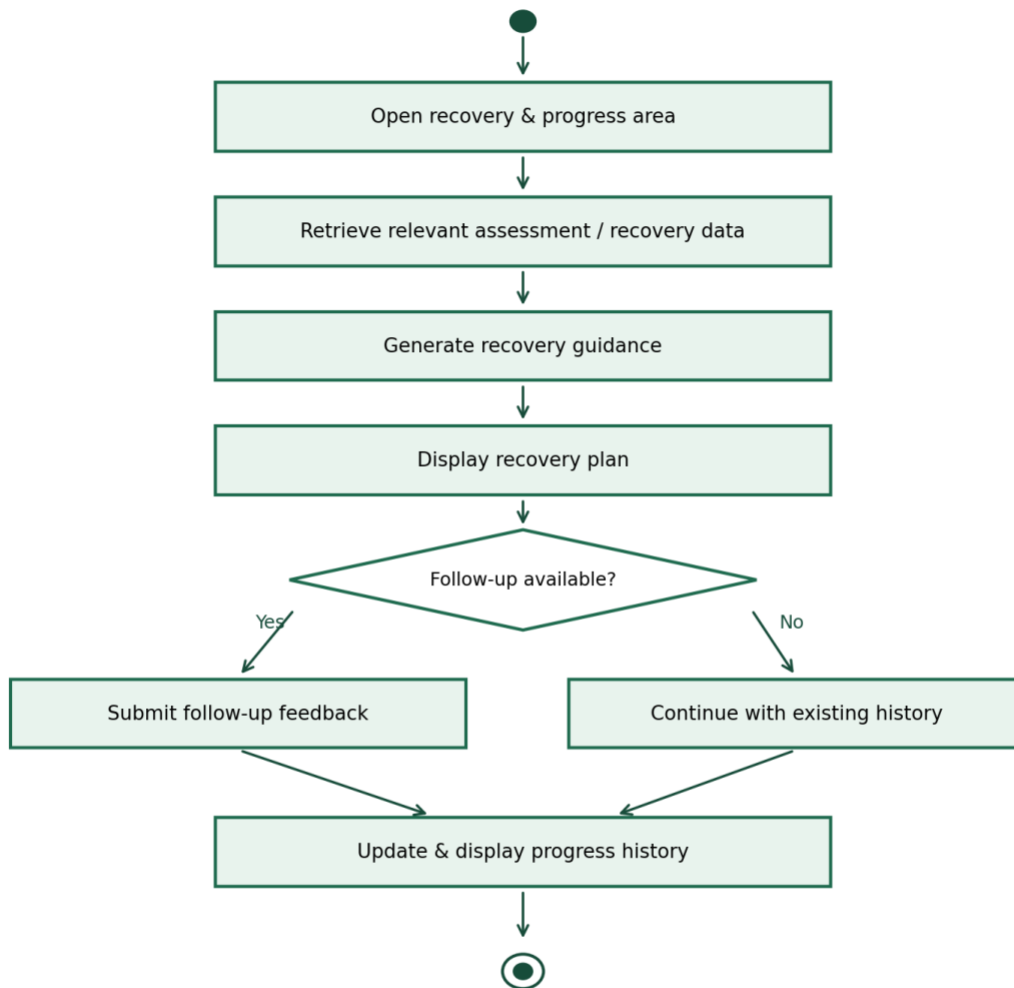


Figure 7. UC-04 Manage Recovery & Progress - detailed use case diagram

Activity Diagram

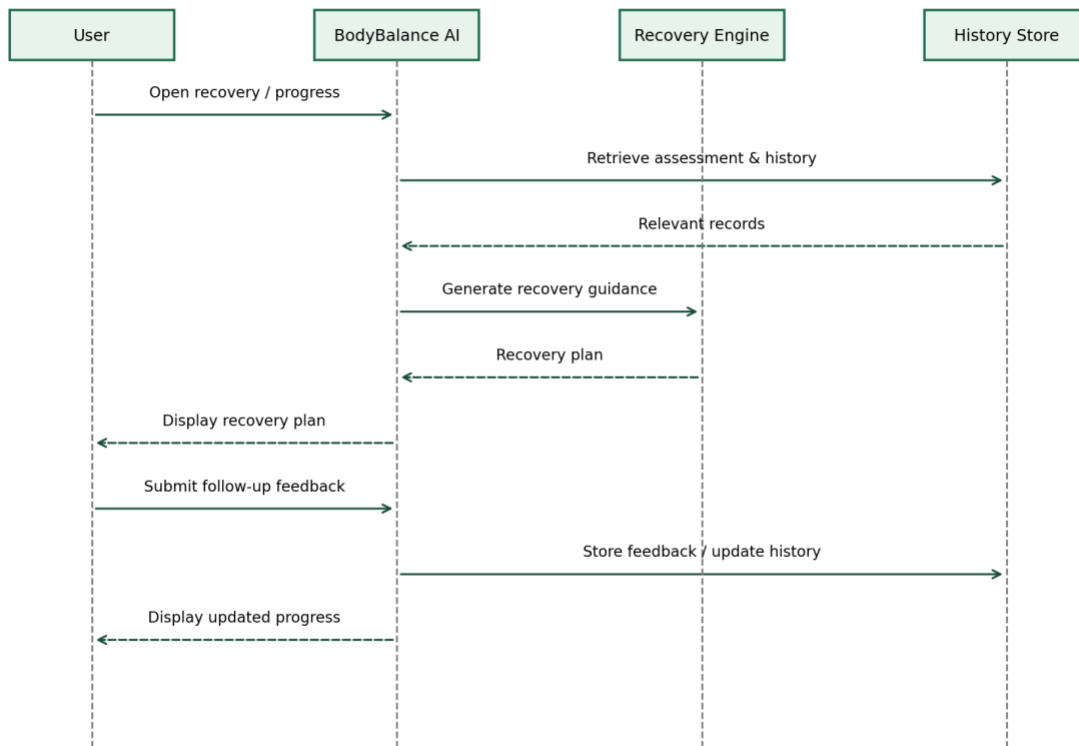
UC-04 Activity Diagram



UC-04 Activity Diagram

Sequence Diagram

UC-04 Sequence Diagram



UC-04 Sequence Diagram

UC-05 Manage Professional Access

Fully Dressed Specification (D2 - Alternate / Authorization Edge)

Field	Content
Use Case	UC-05 Manage Professional Access
Activity Name	Manage Professional Access
Primary Actor	Active Student / Fitness User
Supporting Actors	Fitness Coach / Personal Trainer; Healthcare Professional
Stakeholders & Interests	User: controls what is shared. Fitness Coach: accesses relevant fitness/activity/recovery information. Healthcare Professional: accesses relevant symptom/activity/posture/recovery information.
Preconditions	User is authenticated; professional identity is valid; access is based on user authorization.
Postconditions	Permissions are stored; only authorized information is available to the selected professional; an organized summary may be generated for sharing.
Main Success Scenario	1. User opens professional-access settings. 2. User selects the professional and information to share. 3. System validates and stores permissions. 4. User may generate a professional summary. 5. Authorized professional opens the permitted record. 6. System displays only authorized information.

Extensions / Alternatives	3a. Authorization missing or expired -> access denied. 5a. No permitted records -> display that no authorized information is available.
Special Requirements	Professional access must remain user-controlled and privacy-preserving.
Constraints	Professionals must not receive information outside the user-authorized scope.
Open Issues	Exact authorization duration and revocation workflow require implementation validation.

Detailed Use Case Diagram (D3)

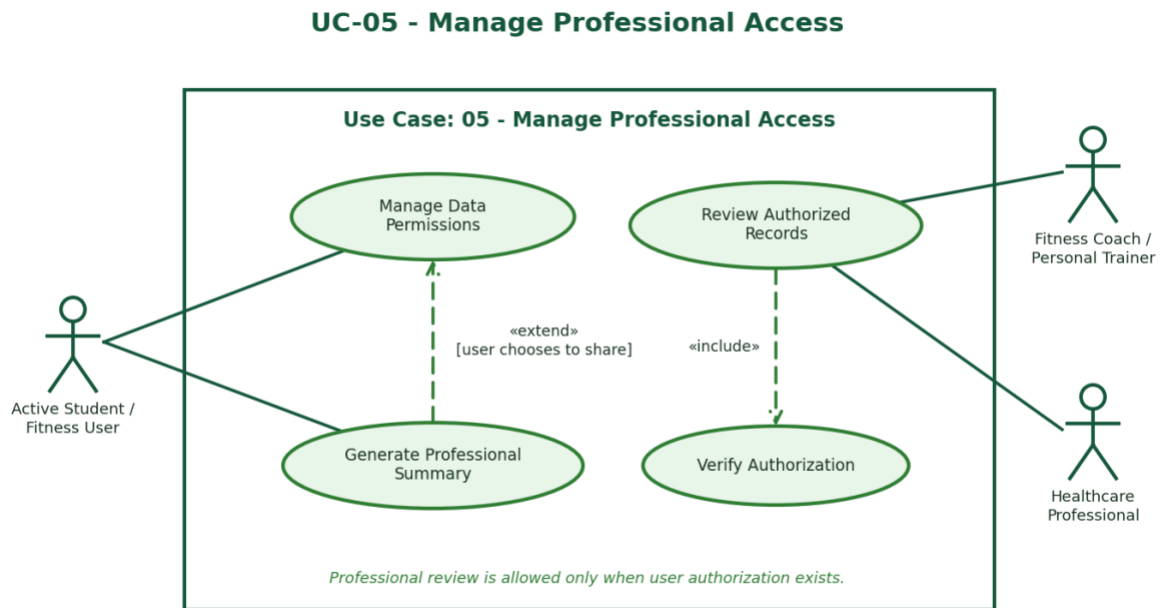
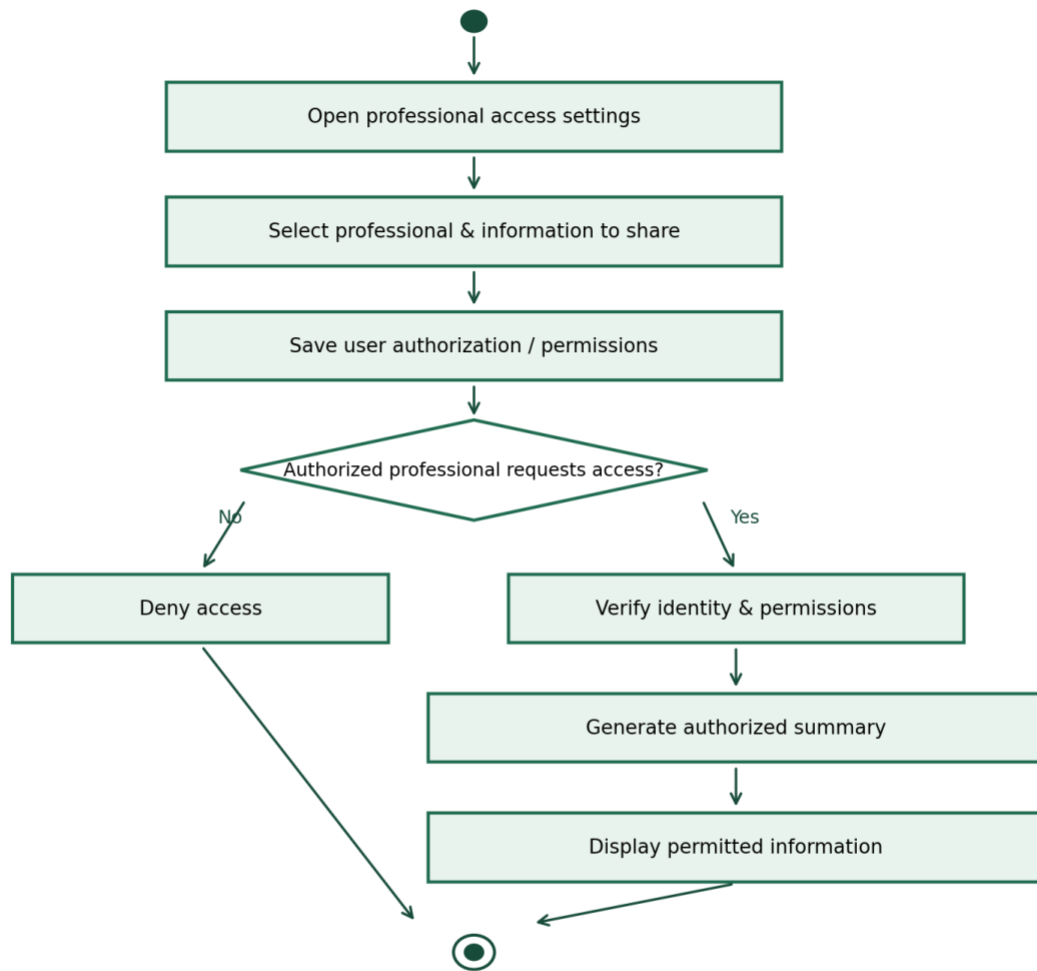


Figure 8. UC-05 Manage Professional Access - detailed use case diagram

Activity Diagram

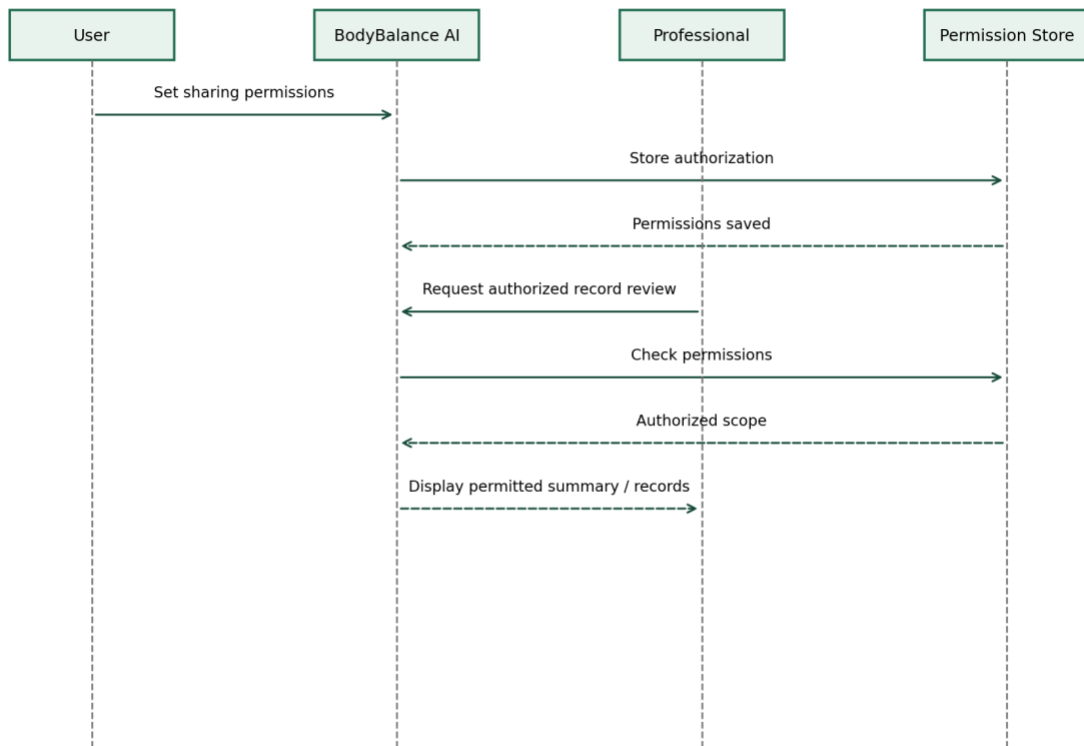
UC-05 Activity Diagram



UC-05 Activity Diagram

Sequence Diagram

UC-05 Sequence Diagram



UC-05 Sequence Diagram

D6. Domain Model (Class Diagram)

The domain model represents problem-domain concepts and relationships rather than database tables or UI components. Multiplicities show how the core BodyBalance AI concepts relate.

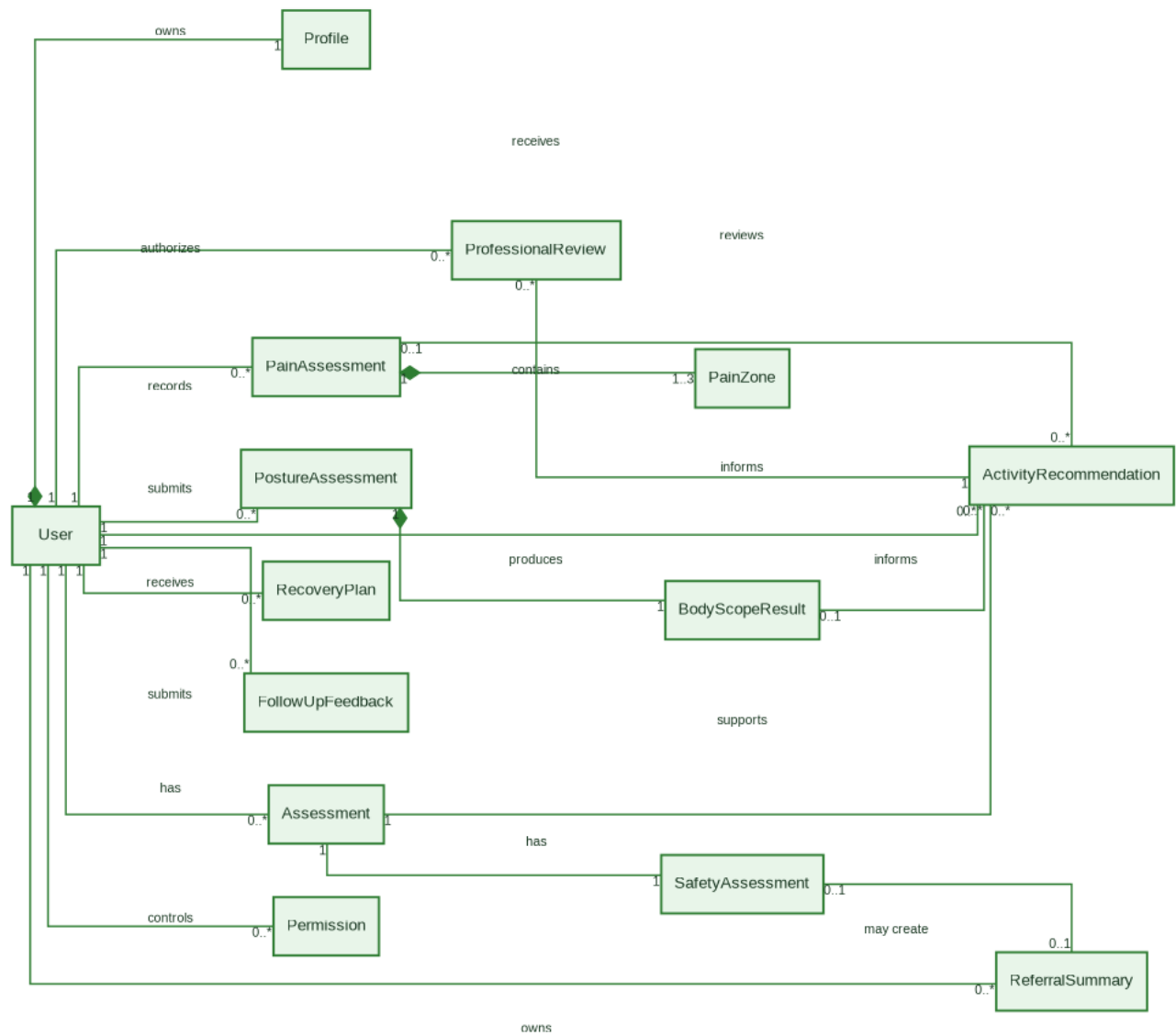


Figure 9. BodyBalance AI Domain Model

D7. Consistency Checklist and Peer Review

D7.1 Consistency Checklist

Consistency Check	Result	Evidence
Every use case is traceable to an actor and a requirement.	Pass	D1 list and D7.3 matrix provide traceability.
Use case diagrams use correct boundary, actors, associations, include/extend.	Pass	Five detailed diagrams use the lecture's recursive use-case granularity approach.
Include/extend directions are correct.	Pass	Include: base -> included. Extend: extending -> base.
Activity diagrams follow each use case workflow and alternatives.	Pass	UC-01 to UC-05 each include an activity diagram; UC-03 contains the detailed safety branches required for the complex case.
Sequence diagrams align with each use case main success scenario.	Pass	UC-01 to UC-05 each include a sequence diagram whose messages follow the corresponding narrative.
Domain model contains problem-domain concepts and multiplicities.	Pass	Classes and multiplicities are shown; UI/database implementation details are excluded.
Preconditions and postconditions are clear.	Pass	They are documented in the brief/full specifications, not inside UML diagrams.
Terminology is consistent across artifacts.	Pass	The five consolidated use case names are used consistently throughout.

D7.2 Peer-Review Sheet

Peer-Review Item	Status	Reviewer Comment
Actors cover all personas and external roles	✓	All identified personas and external system actors are represented appropriately.
Use case names are goal-level and unambiguous	✓	Use case names are clear, goal-focused, and consistently defined.
Include / extend relationships are used sparingly and correctly	✓	Include and extend relationships are used only where required and follow correct UML notation.
Activity and sequence diagrams match their corresponding use case narratives	✓	Activity and sequence diagrams are consistent with the corresponding use case flows.
Class diagram avoids UI/database details and multiplicities make sense	✓	The domain model focuses on domain concepts, with appropriate relationships and multiplicities.
Preconditions and postconditions are clear	✓	Preconditions and postconditions are clearly defined and consistent with each use case.
Terminology is consistent across all artifacts	✓	Terminology and use case naming are consistent throughout all UML artifacts
Diagrams are readable and professionally organized	✓	Diagrams are clear, readable, and professionally organized.

Reviewer name / team: Group 2 Date: 7 oct 2026

D7.3 Requirements-to-UML Traceability Matrix

Lab 3 User Story	Requirements	Lab 4 Use Case	Domain Concepts
US1 Registration & Profile	REQ-1-REQ-3	UC-01	User, Profile
US2 Mental/Physical Check-In	REQ-4-REQ-6	UC-02	Assessment
US3 Pain Locator	REQ-7-REQ-9	UC-02	PainAssessment, PainZone

US4 BodyScope AI	REQ-10-REQ-13	UC-02	PostureAssessment, BodyScopeResult
US5 Activity Recommendation	REQ-14-REQ-16	UC-03	ActivityRecommendation
US6 Recovery Plan	REQ-17-REQ-19	UC-04	RecoveryPlan
US7 Body Timeline	REQ-20-REQ-21	UC-04	Assessment, PainAssessment, BodyScopeResult, RecoveryPlan
US8 Follow-Up Feedback	REQ-22-REQ-23	UC-04	FollowUpFeedback
US9 Professional Referral	REQ-24-REQ-25	UC-03 / UC-05	ReferralSummary
US10 Safety Escalation	REQ-26-REQ-27	UC-03	SafetyAssessment, ReferralSummary
US11 Coach Access	REQ-28-REQ-29	UC-05	User, Profile, Assessment, Permission
US12 Healthcare Professional Access	REQ-30-REQ-31	UC-05	User, Assessment, ReferralSummary, Permission

References

- Elkafrawy, P. (Fall 2026). Software Engineering Lab: From Personas & Requirements to Use Cases & UML. CS3151 Software Engineering course handout.
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- BodyBalance AI project materials: Labs 1-3 and Feasibility Study (personas, requirements, user stories, safety constraints, and project scope).
- Object Management Group. (2017). OMG Unified Modeling Language (OMG UML), Version 2.5.1.
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